Week 3 **Lab Experiment 3**

3] Perform the following DB operations using Cassandra.

a) Create a keyspace by name Library
b) Create a column family by name Library-Info with attributes
   Stud_Id Primary Key, Counter_value of type Counter, Stud_Name,
   Book-Name, Book-Id,
   Date_of_issue
c) Insert the values into the table in batch
d) Display the details of the table created and increase the value of the counter
e) Write a query to show that a student with id 112 has taken a book "BDA" 2 times.
f) Export the created column to a csv file
g) Import a given csv dataset from local file system into Cassandra column family

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Use HELP for help.
cqlsh> DESCRIBE KEYSPACES;

system       system_distributed  system_traces  system_virtual_schema
system_auth  system_schema       system_views

cqlsh> CREATE KEYSPACE Library
   ... WITH replication = {'class': 'SimpleStrategy', 'replication_factor': 1};
cqlsh> USE Library;
cqlsh:library> CREATE TABLE Library_Info (
           ...     Stud_Id int,
           ...     Stud_Name text,
           ...     Book_Name text,
           ...     Book_Id int,
           ...     Date_of_issue date,
           ...     PRIMARY KEY (Stud_Id, Book_Id)
           ... );
cqlsh:library> CREATE TABLE Book_Count (
           ...     Stud_Id int,
           ...     Book_Name text,
           ...     Counter_value counter,
           ...     PRIMARY KEY (Stud_Id, Book_Name)
           ... );
cqlsh:library> BEGIN BATCH
           ... INSERT INTO Library_Info (Stud_Id, Stud_Name, Book_Name, Book_Id, Date_of_issue)
           ... VALUES (112, 'Rahul', 'BDA', 1, '2026-02-24');
           ...
           ... INSERT INTO Library_Info (Stud_Id, Stud_Name, Book_Name, Book_Id, Date_of_issue)
           ... VALUES (113, 'Anita', 'ML', 2, '2026-02-24');
           ... APPLY BATCH;
cqlsh:library> UPDATE Book_Count
           ... SET Counter_value = Counter_value + 1
           ... WHERE Stud_Id = 112 AND Book_Name = 'BDA';
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            ... );
cqlsh:library> CREATE TABLE Book_Count (
           ...     Stud_Id int,
           ...     Book_Name text,
           ...     Counter_value counter,
           ...     PRIMARY KEY (Stud_Id, Book_Name)
           ... );
cqlsh:library> BEGIN BATCH
           ... INSERT INTO Library_Info (Stud_Id, Stud_Name, Book_Name, Book_Id, Date_of_issue)
           ... VALUES (112, 'Rahul', 'BDA', 1, '2026-02-24');
           ...
           ... INSERT INTO Library_Info (Stud_Id, Stud_Name, Book_Name, Book_Id, Date_of_issue)
           ... VALUES (113, 'Anita', 'ML', 2, '2026-02-24');
           ... APPLY BATCH;
cqlsh:library> UPDATE Book_Count
           ... SET Counter_value = Counter_value + 1
           ... WHERE Stud_Id = 112 AND Book_Name = 'BDA';
cqlsh:library>
cqlsh:library> UPDATE Book_Count
           ... SET Counter_value = Counter_value + 1
           ... WHERE Stud_Id = 112 AND Book_Name = 'BDA';
cqlsh:library> SELECT * FROM Library_Info;

 stud_id | book_id | book_name | date_of_issue | stud_name
---------+---------+-----------+---------------+-----------
     113 |       2 |        ML |    2026-02-24 |     Anita
     112 |       1 |       BDA |    2026-02-24 |     Rahul

(2 rows)
cqlsh:library> SELECT * FROM Book_Count;

 stud_id | book_name | counter_value
---------+-----------+---------------
     112 |       BDA |             2
```

```
     112 |       BDA |             2

(1 rows)
cqlsh:library> UPDATE Book_Count
           ... SET Counter_value = Counter_value + 1
           ... WHERE Stud_Id = 112 AND Book_Name = 'BDA';
cqlsh:library> SELECT Counter_value
           ... FROM Book_Count
           ... WHERE Stud_Id = 112 AND Book_Name = 'BDA';

 counter_value
---------------
             3

(1 rows)
cqlsh:library> exit

C:\Users\BPC\Downloads\apache-cassandra-4.1.10-bin\apache-cassandra-4.1.10\bin>docker exec -it my-cassandra cqlsh -e "COPY Library.Library_Info
TO '/tmp/library.csv' WITH HEADER = true;"
Using 7 child processes

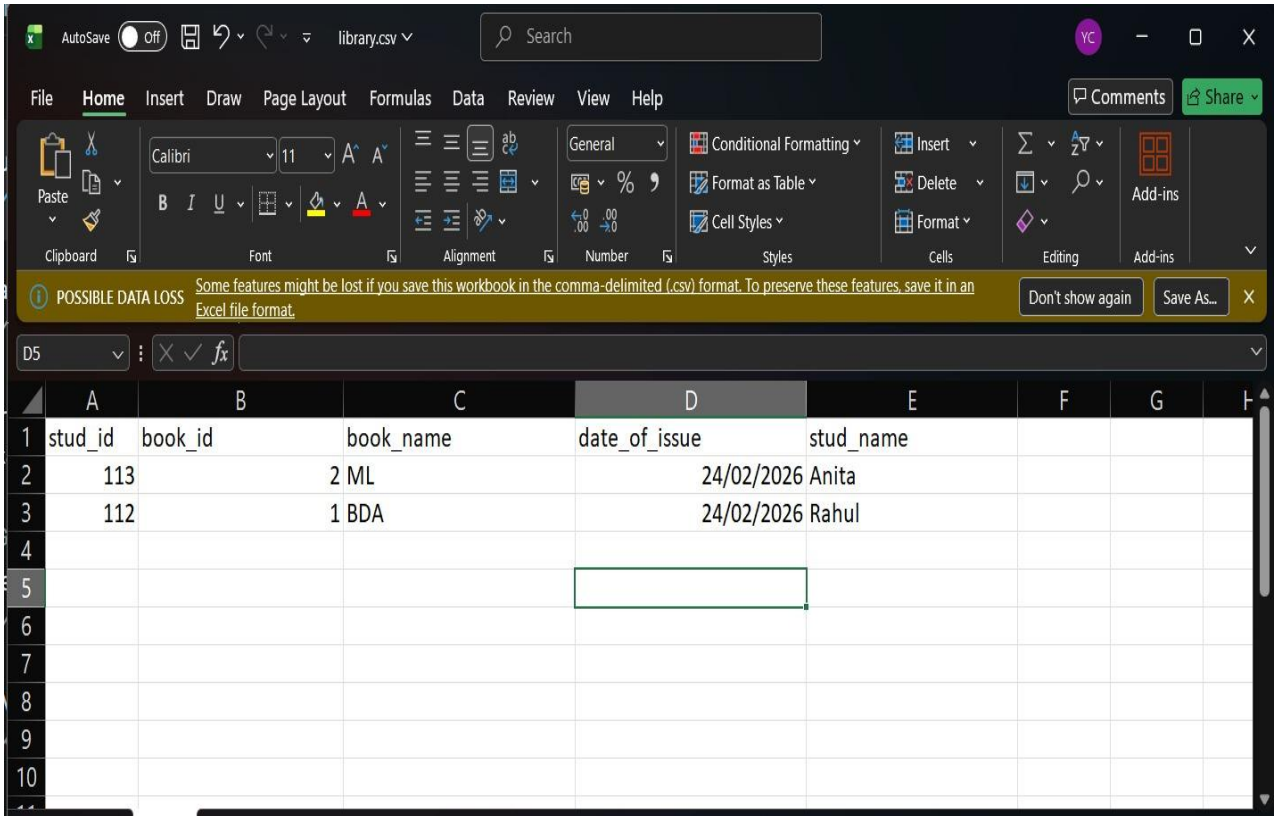
Starting copy of library.library_info with columns [stud_id, book_id, book_name, date_of_issue, stud_name].
Processed: 2 rows; Rate:       7 rows/s; Avg. rate:       3 rows/s
2 rows exported to 1 files in 0.633 seconds.

C:\Users\BPC\Downloads\apache-cassandra-4.1.10-bin\apache-cassandra-4.1.10\bin>docker cp my-cassandra:/tmp/library.csv library.csv
Successfully copied 2.05kB to C:\Users\BPC\Downloads\apache-cassandra-4.1.10-bin\apache-cassandra-4.1.10\bin\library.csv

C:\Users\BPC\Downloads\apache-cassandra-4.1.10-bin\apache-cassandra-4.1.10\bin>docker cp library.csv my-cassandra:/tmp/library.csv
Successfully copied 2.05kB to my-cassandra:/tmp/library.csv

C:\Users\BPC\Downloads\apache-cassandra-4.1.10-bin\apache-cassandra-4.1.10\bin>docker exec -it my-cassandra cqlsh -e "COPY Library.Library_Info
FROM '/tmp/library.csv' WITH HEADER = true;"
Using 7 child processes

Starting copy of library.library_info with columns [stud_id, book_id, book_name, date_of_issue, stud_name].
Processed: 2 rows; Rate:       4 rows/s; Avg. rate:       5 rows/s
2 rows imported from 1 files in 0.379 seconds (0 skipped).
```



stud_id	book_id	book_name	date_of_issue	stud_name
113	2	ML	24/02/2026	Anita
112	1	BDA	24/02/2026	Rahul